

Motion Analysis and Reconstruction of Human Joint Regions for Sparse RGBD Images

Tianzhen Dong*

Yuntao Bai†

Qing Zhang‡

Yi Zhang§

School of Computer Science & Information Engineering
Shanghai Institute of Technology, China

ABSTRACT

We propose a motion deformation analysis framework that can reconstruct the joint region of dynamic human only depending on extremely sparse RGBD images. In the framework, we analyze the deformation mechanism of the joint region from the perspective of its internal material. The non-rigid deformation of a joint region is transformed into the elastic deformation of multiple elements by introducing Finite Element Method. Considering the influence of joint position on the motion deformation analysis, we optimize the joint positions obtained from skeleton tracking prior. Experimental results verify the rationality and effectiveness of our framework.

Index Terms: Human-centered computing—Visualization—Visualization techniques—Treemaps; Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

The growing markets of virtual reality (VR) and augmented reality (AR), combined with the wide availability of RGBD camera devices, are perpetually increasing the demand for reconstruction of dynamic scenes, especially for human-centered non-rigid reconstruction. Recently, most researchers focus on motion deformation of the full body shape, but it is lack of the analysis of deformation mechanism in the region around the joint. It makes the joint region that originally produces strong non-rigid deformation excessively smooth and lack details after fusing continuous RGBD observation data. Studies show that the deformation of body shape is driven by the deformation of several joint regions. The motion reconstruction of joint regions is the core of human motion reconstruction, and its essence is to solve the non-rigid deformation of joint regions.

We propose a motion deformation analysis framework for reconstructing joint regions during human motion. Fig.1 shows the overview of our framework. The joint region at a certain body posture is input into the framework. Its continuous non-rigid deformation is discretized into the elastic deformation of multiple elements by introducing Finite Element Method (FEM) [1, 3]. Based on the analysis of deformation distribution of joint regions, we fit the elastic deformation of all elements and fuse them into the joint region at another body posture according to the joint motion angle. The above process does not focus on the processing of RGBD observation data, which means that our framework can reconstruct the joint regions at different body postures in the case of extremely sparse RGBD images, or even a single RGBD image. Considering the influence of joint position on the motion deformation analysis, we adopt particle swarm strategy to optimize the joint positions obtained from skeleton tracking prior.

*e-mail: dongtianzhen@126.com

†e-mail: ytb_shanghai@163.com

‡e-mail: zhangqing0329@gmail.com

§e-mail: aristormir@163.com

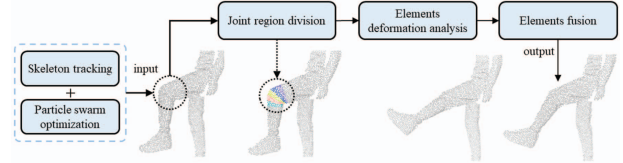


Figure 1: The overview of our framework. 1) The joint region obtained according to the joint position is used as the input of the framework. 2) The inputted joint region is divided into multiple elements; 3) The elastic deformation of each element is analyzed according to the distribution of elastic modulus; 4) All elements are fused into a new joint region as the output of the framework.

2 METHOD OVERVIEW

2.1 Deformation Analysis

We take a knee joint as research object to illustrate the framework. According to our observation, the deformation of joint regions mainly depends on the tangential deformation, which is much stronger than the axial and radial deformations, and has uneven spatial distribution. We believe that the uneven deformation distribution of a joint region is caused by the uneven distribution of internal elastic modulus. We advise that the change of tangential elastic modulus is described by the normal distribution, while the axial and radial elastic moduli are considered as tending to infinity. It means that we do not need to consider the axial and radial deformations in the next the analysis for elements.

Given the joint region composed of the point cloud in the region around knee joint. Point cloud can be obtained from RGBD images taken by a single RGBD camera. Geometrically, we can divide the joint region into multiple cylinder-like objects along the tangential direction that has the strongest deformation. Each cylinder-like object represents an element that contains a series of 3D points and two shape attributes, geometric center and section radius. And all elements constitute an element set, $\mathcal{J} = \{S_0, S_1, \dots, S_n\}$. When the number of elements is sufficient, the shape of each element is regarded as a cylinder and its internal elastic modulus is considered as uniform in the same direction (axial, radial and tangential). It means that elastic deformation of an element is also uniform in the same direction. Therefore, the continuous non-rigid deformation of the joint region is discretized into elastic deformation of multiple elements. At the same time, the stress model of an element can be simplified to that of the rectangular element [1, 3], as shown in the Fig.2(a). The nodal positions in the rectangular element can be obtained according to geometric center and section radius of the element in element set. For element $S_i \in \mathcal{J}$, its internal tangential elastic modulus is expressed as follows:

$$E_t(S_i) = 1 - \exp \frac{\|c_i - j_k\|^2}{2d^2} \quad (1)$$

Where, $c_i \in \mathbb{R}^3$ is the geometric center of element S_i , $j_k \in \mathbb{R}^3$ is knee joint position. We generally set parameter $d = 7$ to control

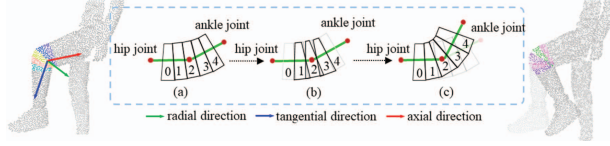


Figure 2: 2D diagram of element deformation and fusion.

the difference of elastic modulus between two adjacent elements. According to the Hooke's law and the equivalent nodal force [1], the displacement of nodes in the rectangular element S_i^r is: $L_r(S_i^r) = kF/E_t(S_i^r)$. We set $k = 0.25$ to limit the maximum deformation of S_i^r . Because it is unrealistic to directly measure the force on the joint region, joint motion angle θ is considered equivalent to force value F : $\theta \Leftrightarrow F$. As shown in the Fig.2(b), the deformation of rectangular elements represents the tangential elastic deformation of the elements in element set. A series of 3D points of S_i are translated to follow its elastic deformation. Their translation can be obtained according to the shape function of the rectangular element S_i^r . For more details about the stress model of rectangular element and shape function, please refer to [1, 3].

2.2 Element Fusion

The different elastic deformations of elements lead to the joint region may be discontinuous at any place. We set a rigidity pose transformation for each element to fuse a complete joint region. The pose parameter is a rotation matrix $R(\alpha, c_i) \in \mathbb{R}^{3 \times 3}$ that represents a rotation of α degrees around an axis passing through c_i (the geometric center of element S_i) and parallel to the axial direction. Because the deformation of an element mainly depends on tangential elastic deformation and because transformation of an element can affect each other in FEM, the pose transformation of each element is designed as an incremental transformation and is defined as follows:

$$S_i^* = R_0 R_1 \dots R_i S_i \quad (2)$$

Where, $R_j = R((1 - E_t(S_j))\theta, c_j)$, $0 \leq j \leq i$. A series of 3D points of S_i are transformed to follow its pose transformation. The incremental transformation makes the orientation and position of S_i approximately continuous among elements, as shown in the Fig.2(c). All elements perform transformation in order from S_0 to S_n .

2.3 Joint Location

Some advanced human skeleton tracking approaches often produce a phenomenon that joint position deviates from bodily point cloud due to motion blur and marginal false depth value, resulting in the wrong or incomplete joint region is input into the framework. However, the performance of these approaches has the similar movement trend as real human motion and the tracked joint positions are often around the bodily point cloud. We uniformly sample many particles in the local space around tracked joint position. If a particle meets the following conditions: its projection position in the 2D image domain is located in the human foreground region; it cannot be at the edge of the bodily point cloud, its position is used as the optimized joint position.

3 EXPERIMENTS

We use a RGBD camera to record RGBD image sequences of joint regions during human motion, and take it as the real dataset. We evaluated the framework in the real dataset. The point cloud obtained from back-projection of the RGBD image recorded at a certain time in real dataset as the ground truth at this time. The ground truth at a certain time is used as the input of our framework. The number of elements is 64. Fig.3 show that the comparison of our output with ground truth at another time and SobolevFusion [2]. All models are

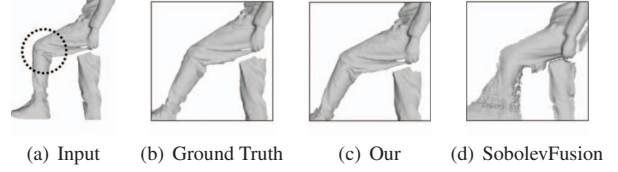


Figure 3: Comparison between our output with ground truth and SobolevFusion. We reconstructed the knee joint region during the kick sequence. (a) is the inputted leg shape. (b), (c) and (d) are close-ups of the knee joint region.



Figure 4: More results of knee joint region reconstruction.

displayed in meshing. (b) shows the natural deformation captured by the RGBD camera. (c) shows the deformation solved by our framework under the condition that only a single RGBD image is used. (d) shows the result of non-rigid tracking of SobolevFusion frame by frame. Compared with SobolevFusion, our output has a deformation distribution more similar to ground truth and a more refined shape. And (c) also shows that SobolevFusion lacks robustness for large deformation between frames. Fig.4 shows the more results of the framework using Fig.3(a) as input.

4 CONCLUSION

In this paper, we propose a motion deformation analysis framework for reconstructing joint regions during human motion. We analyze the deformation mechanism of a joint region from the perspective of material properties and use the elastic deformation of multiple elements to fuse the natural non-rigid deformation of the joint region without using a large number of RGBD observation data for non-rigid tracking. Experiments verify that the established deformation analysis framework based on FEM is applicable. In the reconstruction and simulation of living creatures with articulation structures, we provide a framework to analyze the non-rigid deformation of the region around articulation from the physical deformation mechanism, instead of relying on RGBD observation data. It is critical for non-rigid object's motion reconstruction and simulation.

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